# ANALISI FUNZIONALE

$f \in \mathcal{C}^1([a,b])$

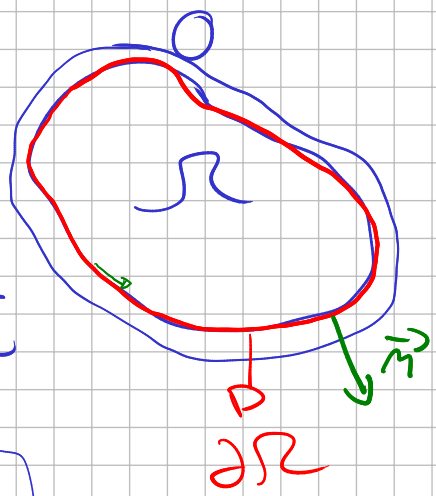
$$\rightarrow \underbrace{\int_a^b f'(x)\,dx} = \underbrace{f(b)} - \underbrace{f(a)}$$

$\Omega \subset \mathbb{R}^d$

$\partial\Omega$ LISCIO

$u: \Omega \to \mathbb{R}$ $\quad \mathcal{C}^1(O)$ $\quad O \in \bar{\Omega}$

APERTO

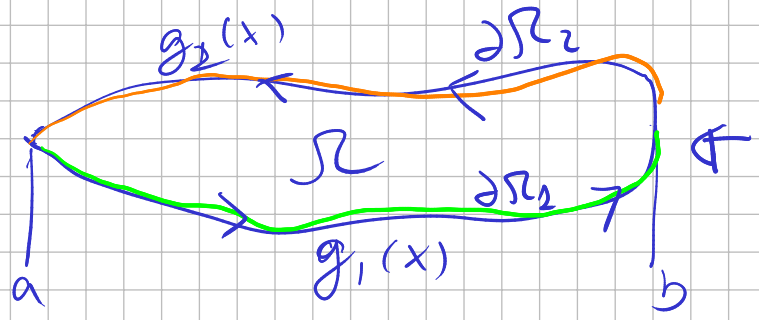



$$\int_\Omega \partial_{x_i} u(x)\,dx = \boxed{\int_{\partial\Omega} u(x)\cdot n_i(x)\,dS}$$

$n$ NORMALE A $\partial\Omega$ USCENTE DA $\Omega$

**DIM** $\Omega$ DOMINIO SEMPLICE 2D

$$\Omega = \{(x,y) : x \in [a,b],\ g_1(x) < y < g_2(x)\}$$




$i=2$

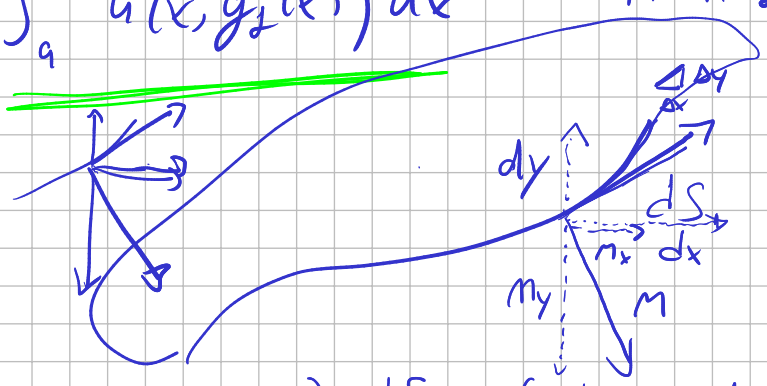
$\boxed{x_i = y}$

$$\int_\Omega \partial_y u(x,y)\,dx\,dy = \underbrace{\int_a^b \underbrace{\int_{g_1(x)}^{g_2(x)} \partial_y u(x,y)\,dy}_{1D}\,dx}$$

$$= \int_a^b \left[u(x,g_2(x)) - u(x,g_1(x))\right] dx$$

$$= \int_a^b u(x, g_2(x))\,dx - \int_a^b u(x, g_1(x))\,dx$$

$\|\vec{n}\| = 1$




$$\int_{\partial\Omega} u(x,y)\cdot n_y \, dS$$

$$= \int_{\partial\Omega_1} u(x,y)\cdot n_y \, dS$$

$$+ \int_{\partial\Omega_2} u(x,y)\cdot n_y \, dS$$

$$\vec{n}\cdot dS = (dy, -dx)$$

$$(n_x, n_y)\, dS = (dy, -dx)$$

$$n_y\, dS = -dx$$

$$= -\int_a^b u(x, g_1(x))\,dx - \int_b^a u(x, g_2(x))\,dx$$

$$= -\int_a^b u(x, g_1(x))\,dx + \int_a^b u(x, g_2(x))\,dx$$

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TEO DIVERGENZA

$$u = \begin{pmatrix} u_1 \\ \vdots \\ u_d \end{pmatrix} \qquad u_i \in C^1(O)$$

$$\int_\Omega \operatorname{div}(\vec{u})\, d\vec{x} = \int_\Omega \sum_{i=1}^d \partial_{x_i} u_i \, d\vec{x}$$

$$= \int_{\partial\Omega} \sum_{i=1}^d u_i \cdot n_i \, dS = \int_{\partial\Omega} \vec{u}\cdot\vec{n}\, dS$$

$$f \in C^1(\Omega) \qquad \vec{u} \in (C^1(\Omega))^d$$

$$\int_{\partial\Omega} f\cdot\vec{u}\cdot\vec{n}\, dS = \int_{\partial\Omega} \operatorname{div}(f\cdot\vec{u})\, dx$$

$$= \int_\Omega f\cdot \operatorname{div}(\vec{u})\, dx + \int_\Omega \nabla f \cdot \vec{u}\, dx$$

$\boxtimes$

# SPAZI FUNZIONALI $\mathbb{R}^n$, $\mathbb{C}^n$

$\mathcal{C}(\Omega)$, $\quad \|f\|_\infty = \sup_{x\in\Omega} |f(x)|$

$\mathcal{C}^k(\Omega)$ $\quad \|f\|_{\mathcal{C}^k} := \sum_{p_1+\dots p_d \le k} \|\partial_{x_1}^{p_1} \dots \partial_{x_d}^{p_d} f\|_\infty$

$\mathcal{C}^\infty(\Omega)$ $\quad \|f\|_\infty$

$$L^2(\Omega) := \left\{ f : \Omega \to \mathbb{R} \;\middle|\; \int_\Omega |f|^2\, dx < \infty \right\} \Big/ \sim$$

$$\|f\|_{L^2} = \|f\|_2 := \sqrt{\int_\Omega |f|^2\, dx}$$

$$f(x) = \begin{cases} 3 & x = 0 \\ 0 & x \neq 0 \end{cases} \qquad \Omega = (-1,1)$$

$$\int_{-1}^{1} |f|^2\, dx = 0 \qquad f \not\equiv 0$$

$f \sim g \qquad f(x) = g(x)$ q.o.

$$\int_\Omega |f(x) - g(x)|^2\, dx = 0$$

$\Rightarrow$ NORMA ✓

es. $\mathbb{R}^2$ $\quad (1\ 1) \quad L\begin{pmatrix} x \\ y \end{pmatrix} := (1\ 1)\begin{pmatrix} x \\ y \end{pmatrix} = x + y$

$V$ SPAZIO FUNZIONALE

$L : V \to \mathbb{R}$ FUNZIONALE LINEARE

$$\underline{L(\alpha u + \beta v) = \alpha L(u) + \beta L(v)} \qquad u, v \in V, \quad \alpha, \beta \in \mathbb{R}$$

es 1

$L_{x_0} : \mathcal{C}(\Omega) \to \mathbb{R}$ $\quad L_{x_0}(\underbrace{u}_{\in V = \mathcal{C}(\Omega)}) = \underbrace{u(x_0)}_{} \in \mathbb{R}$

$$L_{x_0}(\alpha u + \beta v) = \alpha u(x_0) + \beta v(x_0)$$

es 2 $\quad L : L^1(\Omega) \to \mathbb{R}$

$$L(u) := \int_\Omega u(x)\,dx \in \mathbb{R}$$

$$L(\alpha u + \beta v) = \int \alpha u + \beta v \; dx = \alpha \int_\Omega u(x)\,dx + \beta \int_\Omega v(x)\,dx =$$

$$= \alpha L(u) + \beta L(v)$$

$V$, SPAZIO NORMATO

$L$ LIMITATO se $\exists\, C > 0$ : $|L(u)| \le C \cdot \|u\|_V \quad \forall u \in V$.

se $V$ BANACH (NORMATO e COMPLETO) $L$ LIMITATO $\Rightarrow$ $L$ CONTINUO

es 1 $\quad L_{x_0}(u) := u(x_0)$ IN $V = \mathcal{C}^0(\Omega)$ CON $\|\cdot\|_\infty$

$$|L_{x_0}(u)| = |u(x_0)| \le \sup_{x \in \Omega} |u(x)| = \|u\|_\infty$$

CON $C = 1$

$\Downarrow$

LIMITATO ✓

CONTROESEMPIO NON LIMITATO

$L : \mathcal{C}^\infty([0,1]) \to \mathbb{R} \quad L(u) = u'(0) \qquad V, \|\cdot\|_\infty$

$$\|u\|_\infty := \sup_{x \in \Omega} |u(x)|$$

$u_n : \|u_n\|_\infty \le 1$

$$|L(u_n)| \xrightarrow[n \to \infty]{} \infty \quad \Rightarrow \exists\, c : \underbrace{|L(u_n)|}_{\infty} \le c \underbrace{\|u_n\|_\infty}_{\le 1}$$

$u_n(x) := \sin(nx)$

$\Rightarrow \|u_n\|_\infty = 1 \quad \forall n$

$L(u_n) = u_n'(0) = n\cos(nx)\big|_{x=0} = n$

$|L(u_n)| \to \infty$ $\boxtimes$

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<u>SPAZIO DUALE</u> di $V$ = SPAZIO DI TUTTI I FUNZIONALI LINEARI LIMITATI SU $V$

$$V^* = V' := \{ F : V \to \mathbb{R} \mid \text{LINEARI E LIMITATI} \}$$

$\forall L \in V^*$

- $$\|L\|_{V^*} := \sup_{\substack{\|u\|_V \le 1 \\ u \in V}} |L(u)| = \sup_{\|u\|_V \neq 0} \frac{|L(u)|}{\|u\|_V}$$

$\mathbb{R}^2 \quad L\begin{pmatrix} x \\ y \end{pmatrix} = x+y = \underbrace{(1\ 1)}\begin{pmatrix} x \\ y \end{pmatrix}$

$(\mathbb{R}^2)^* = \{ (\alpha\ \beta) \in (\mathbb{R}^2)^T \}$

$\mathbb{R}^2 \qquad L_{(\alpha\ \beta)}\begin{pmatrix} x \\ y \end{pmatrix} := (\alpha\ \beta)\begin{pmatrix} x \\ y \end{pmatrix}$

$= \alpha x + \beta y$

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$H$ <u>SPAZIO DI HILBERT</u> CON UN PRODOTTO SCALARE $\langle f, g \rangle$

$\Rightarrow$ NORMA INDOTTA DAL PR. SC. $\|f\| = \sqrt{\langle f, f \rangle}$

SE $V$ È COMPLETO RISPETTO A

$\Rightarrow$ È UNO SPAZIO DI HILBERT

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$L^2(\Omega) \qquad (f, g)_{L^2} := \int_\Omega f(x) \cdot g(x)\, dx$

HILBERT ☺

$C^0(\Omega)$ $\|\ \|_\infty$ NON È UNO SP. HILBERT

PERCHÉ NON È COMPLETO $\quad x^n \in C^0([0,1])$

$$x^n \xrightarrow[n\to\infty]{} \begin{cases} 0 & 0 \le x < 1 \\ 1 & x = 1 \end{cases} \notin C^0([0,1])$$

$(u,u) \ge 0$

$(u,u) = 0 \iff u \equiv 0$

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TEOREMA DI RAPP. DI RIESZ

$H$ HILBERT $\quad L \in H^* \quad \exists\, v_L \in H$

$$\underbrace{L(u)} = (u, v_L)_H \qquad \forall\, u \in H.$$

e VICEVERSA $\forall\, v \in H \Rightarrow \exists\, L_v \in H^* : L_v(u) = (u,v)_H$.

$\|L\|_{H^*} = \|v_L\|_H \qquad \Downarrow \qquad \|L_v\|_{H^*} = \|v\|_H$

es.

$L^2([0,1]) = V = H$

$\underline{L(u)} := \underbrace{\int_0^1 u(x)\, dx}$ $\qquad v_L(x) := \underline{1} \in L^2([0,1]) = H$

$(v_L, u) = \int_0^1 1 \cdot u(x)\, dx = L(u)$

$H = H^*$

$\Downarrow$

BIEZIONE

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FUNZIONALE BILINEARE SU $V$

$a : V \times V \to \mathbb{R}$

- **BILINEARE** $\quad a(\alpha u + \beta w, v) = \alpha a(u,v) + \beta a(w,v)$

$$\forall u, w, v \in V \qquad \forall \alpha, \beta \in \mathbb{R}$$

- **CONTINUO** $\quad \exists M > 0 : \quad a(u,v) \leq M \|u\|_V \|v\|_V \quad \forall u, v \in V$

- **SIMMETRICO** $\quad a(u,v) = a(v,u) \quad \forall u, v \in V$

- **POSITIVO** $\quad a(u,u) \geq 0 \quad \forall u \in V$

  $a(u,u) > 0 \quad \forall u \neq 0$

- **COERCITIVO** / **COERCIVO** $\quad \exists \alpha > 0 : \quad a(u,u) > \alpha \|u\|_V^2 \quad \forall v \in V.$

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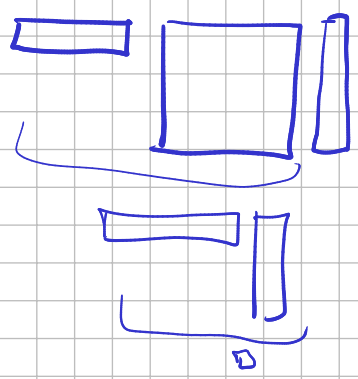
es. IN $\mathbb{R}^d$ $\quad$ LE FORME BILINEARI SONO LE MATRICI $A \in \mathbb{R}^{d \times d}$

$u, v \in \mathbb{R}^d$

$$a(u,v) := u^T A v \in \mathbb{R}^d$$



CONTINUA

$\Downarrow$

$$M = \max_{i=1,\dots,d} \lambda_i(A) \qquad \lambda_i \text{ AUTOVALORI}$$

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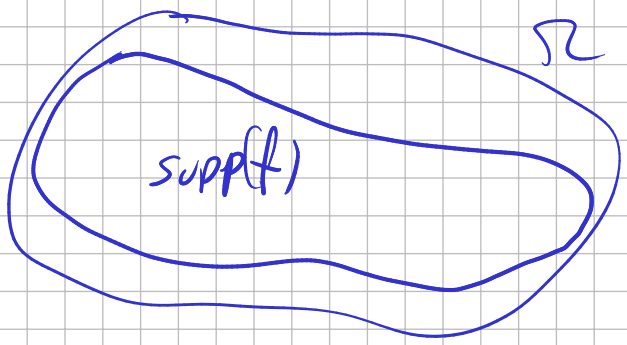
$\Omega \subset \mathbb{R}^d \qquad f : \Omega \to \mathbb{R} \qquad$ supporto di $f$

$$\mathrm{supp}(f) = \overline{\{x \in \Omega : f(x) \neq 0\}}$$

$f$ HA SUPPORTO COMPATTO se $\mathrm{supp}(f)$ è COMPATTO




$\Omega$ APERTO

$D(\Omega) := C_c^\infty(\Omega)$ FUNZIONI $C^\infty$ A SUPPORTO COMPATTO.

• CONVERGENZA IN $D(\Omega)$

$\{f_n\} \subseteq D(\Omega)$ $\quad f_n \to f$ IN $C_c^\infty(\Omega) = D(\Omega)$

se $\exists\, K \subset \Omega$ COMPATTO $\quad \mathrm{supp}(f_n) \subseteq K \quad \forall n$

e se $\quad \partial_{x_1}^{p_1} \dots \partial_{x_d}^{p_d} f_n \longrightarrow \partial_{x_1}^{p_1} \dots \partial_{x_d}^{p_d} f \quad \forall\, p_1, \dots, p_d \in \mathbb{N}^d$

$\Rightarrow f_n$ CONVERGE A $f$

## DISTRIBUZIONI

FUNZ LINEARI $T : \underset{C_c^\infty(\Omega)}{\overset{\cap}{D(\Omega)}} \to \mathbb{R}$

CONTINUE $\quad \lim_{k \to \infty} T(\varphi_k) = T(\varphi)$

$\forall\, \varphi_k \to_D \varphi \in D(\Omega)$

$D^*(\Omega)$ è IL DUALE DI $D(\Omega)$, OD È LO SPAZIO DELLE DISTRIBUZIONI.

ESEMPIO $\quad \delta_a$ DIRAC DELTA

$\delta_a(\phi) = \phi(a) \quad \forall\, \phi \in D(\Omega)$

IN $L^2$ NON FUNZIONA, IN $D(\Omega)$ È UN FUNZIONALE

È UNA DISTRIBUZIONE?

$$\phi_n \to \phi \qquad \lim_{n\to\infty} \sup |\phi_n(x) - \phi(x)| = 0$$

$$|\delta_a(\phi_n) - \delta(\phi)|$$

$$\lim_{n\to\infty} |\phi_n(a) - \phi(a)| = 0$$

DEF

- $\{T_n\}_{n=1}^{\infty} \subseteq D^*(\Omega)$ e $T \in D^*(\Omega)$

$$T_n \to T \quad \text{se} \quad \lim_{n\to\infty} T_n(\phi) = T(\phi) \quad \forall \phi \in D.$$